# RAMAKRISHNAN PANCHACHARAM

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## EDUCATION

**Purdue University** — West Lafayette, IN
*B.S. Electrical Engineering* — Expected May 2029
**New Providence High School** — New Providence, NJ
*High School Diploma, GPA 4.615, AP Scholar with Distinction* — June 2026

## RESEARCH AND EXPERIENCE

**ML2P (Mapping Machine Learning to Physics), DARPA-funded, Stevens Institute of Technology** — Hoboken, NJ
*Undergraduate Researcher, Power Instrumentation (paid)* — Jun 2026 - Present

- Calibrated a **six-rail INA226 current-sense system** on a Raspberry Pi and Pixhawk 6C drone against a Monsoon laboratory power monitor to **1.07% mean agreement** across about 2,200 samples, and built the GPIO handshake that time-aligns the two instruments.
- Wrote the Linux data pipeline for a 54-trial measurement campaign, collecting over **48 million data points**; its trace-level window alignment caught an averaging artifact that had **overstated thermal sensitivity sixfold**.
- Built the isolation protocol behind those trials on the Pixhawk 6C platform: verified rfkill radio cutoffs, pinned camera and CPU-governor state, fixed thermal precondition. Measured **13.1 mW/fps** for camera frame rate.
- Second author on the resulting platform paper, accepted at **IEEE MILCOM 2026**.

**T2D2 (Thornton Tomasetti), AI structural-inspection startup** — New York, NY
*Machine Learning Intern* — Summer 2023

- Built semantic-segmentation models for structural damage detection, reaching **90%+ accuracy** on real inspection imagery.
- Supervised annotation of **25,000+ images**, setting labeling standards with structural engineers.

**NJ Cybersecurity and Communications Integration Cell (NJCCIC)** — New Jersey
*Advanced Cybersecurity Intern and Team Lead* — Jul 2024 - Oct 2024

- **Led 10 interns**; built capture-the-flag challenges and a cyber escape room, and **presented to 300+ middle schoolers** and the NJ Department of Homeland Security. **1st place**, statewide cyberattack-defense simulation.

## ENGINEERING PROJECTS

**Purdue ROV, X-19 Test Bench Power Board** — West Lafayette, IN
*Electrical Team Member* — Sep 2026 - Present

- Designing the **12 V power board** in KiCad for the thruster ESCs, solenoids and Raspberry Pi: XT60, barrel-jack or USB-C PD input with ideal-diode reverse protection and TVS, a **hot-swap controller** on 12 V, and a **12 V to 5 V buck** with a load switch for the Pi.
- Designing **rail monitoring and fault protection**: an STM32 reads **four INA237 monitors** and a temperature sensor, reports over CAN via a **TCAN1044**, and is meant to shut off faulted outputs. MCU and comms sheet drawn.

**Animatronic Show Control System** — Orlando, FL
*2nd, TSA Nationals | 1st, TSA States (x2)* — Jul 2026

- Designed **custom Arduino shield PCBs in KiCad**: schematic capture, board layout, and custom footprint libraries, plus a dedicated **5 V buck-converter board** sized for LED inrush current.
- Wrote the embedded C firmware: a FastLED engine driving **320 WS2812B LEDs** per shield through a five-state scene machine, pulse-synced to a DFPlayer audio module.
- Fabricated and assembled **three scene boards**, reworking power delivery and mechanical mounting between revisions.

**Helios Object-Tracking Turret** — Louisville, KY
*TSA Nationals qualifier* — Jul 2023

- Servo-driven pan/tilt turret tracking objects in real time: **OpenCV vision on a Raspberry Pi**, motion control on an Arduino Uno over serial, split so vision and actuation timing did not contend.

**Reinforcement-Learning Quadruped (PyTorch, MuJoCo)** — New Providence, NJ
*Independent Project* — Jan 2025

- Implemented a Gaussian policy network and policy-gradient training loop **from scratch, without an RL library**, and trained an Ant-v5 quadruped to a stable forward gait.

## LEADERSHIP

**New Jersey Technology Student Association, State Treasurer** — Sep 2025 - Jun 2026

- Managed roughly **$100K in revenue** for a **14,000+ member organization** across 110+ chapters; organized state conferences and represented NJ at Nationals.

**STEM Club, New Providence High School, Vice President** — Sep 2025 - Jun 2026

- Grew TSA Nationals attendance from **~15 to ~60 students** and TEAMS Nationals qualifiers from **2 of 6 to 16 of 18**.

## SKILLS

**Hardware:** KiCad (schematic capture, PCB layout), buck-converter design, INA226/INA237 current sensing, I2C, CAN, STM32 (SWD), power-rail isolation, PID control, Fusion 360
**Test and validation:** measurement-campaign design, instrument calibration against a lab reference, Python data pipelines, measurement-artifact detection
**Software:** Python, C, Embedded C, JavaScript, Linux, Git; PyTorch, TensorFlow, OpenCV, MuJoCo
**Platforms:** Raspberry Pi 5, NVIDIA Jetson Orin NX, Pixhawk 6C, Monsoon power monitor, Arduino